



RV College of Engineering

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Federated Heterogeneous Graph Neural Networks for Trade-Based Money Laundering Detection

MAJOR PROJECT REPORT (CS481P)

Submitted by,

Nikhil Vasu

1RV22CS128

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1RV22CS211

Under the guidance of

Dr. Nagaraja G.S

Sanjana Ravindra Otihal

Professor

Assistant Professor

Dept. of CSE

Dept. of CSE

RV College of Engineering

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In partial fulfillment for the award of degree of

Bachelor of Engineering

in

Computer Science and Engineering

Department of CSE

2025-26





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(Autonomous institution affiliated to VTU, Belagavi)
Department of Computer Science and Engineering



CERTIFICATE

Certified that the major project (CS481P) work titled ***Federated Heterogeneous Graph Neural Networks for Trade-Based Money Laundering Detection*** is carried out by **Nikhil Vasu (1RV22CS128)**, **Rohith Biradar (1RV22CS164)** and **Suraj Chanaveeragoudra (1RV22CS211)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

Guide	Head of the Department	Dean CSE Cluster	Principal
Dr. Nagaraja G.S	Dr. Shanta Rangaswamy	Dr. Ramakanth Kumar P	Dr. K. N. Subramanya

External Viva

Name of Examiners	Signature with Date
1.	
2.	

DECLARATION

We, **Nikhil Vasu, Rohith Biradar** and **Suraj Chanaveeragoudra** students of eighth semester B.E., Department of Computer Science and Engineering, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Federated Heterogeneous Graph Neural Networks for Trade-Based Money Laundering Detection**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Computer Science and Engineering** during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

Name

Signature

1. Nikhil Vasu(1RV22CS128)
2. Rohith Biradar(1RV22CS164)
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ABSTRACT

Trade-Based Money Laundering (TBML) has emerged as a significant challenge within modern financial systems due to the increasing complexity and scale of global trade transactions. Conventional Anti-Money Laundering (AML) systems primarily rely on isolated rule-based monitoring frameworks that are often ineffective in identifying hidden transactional dependencies and multi-hop laundering typologies across interconnected banking networks. To address these limitations, this work developed a privacy-preserving Federated Heterogeneous Graph Neural Network (HGNN)-based framework for real-time multi-class TBML detection across distributed institutional environments.

The proposed framework modeled accounts, trade entities, SWIFT transactions, and trade documents as heterogeneous financial graphs using Memgraph. A hybrid analytical architecture integrating Graph Attention Network (GAT)-based convolutional layers and Long Short-Term Memory (LSTM)-based temporal sequence modeling was employed to capture relational topology and evolving transactional behavior. Federated learning using the Federated Averaging (FedAvg) aggregation strategy enabled collaborative cross-bank model optimization without centralized transactional data sharing. The implemented workflow incorporated Apache Kafka-based transaction streaming, graph construction, federated fraud inference, Suspicious Transaction Report (STR) generation, and regulator-facing compliance synchronization using Python, PyTorch Geometric, FastAPI, PostgreSQL, Memgraph, and React.js.

Experimental evaluation across distributed banking clients demonstrated stable federated convergence with Fraud-class Precision values approaching 0.98, F1-Scores nearing 0.84, and Precision-Recall Area Under Curve (PR-AUC) values exceeding 0.92. Concurrent scalability evaluation additionally maintained dashboard retrieval latency near 156–162 ms and fraud-filtering response latency near 148–159 ms under increasing institutional workloads, thereby validating the operational suitability of the proposed framework for scalable and privacy-preserving TBML detection. The framework can be extended into a fully integrated banking compliance platform with automated risk response, transaction control, and regulatory workflows. These enhancements would transform it from a monitoring system into a complete financial crime prevention ecosystem.



CONTENTS

Abstract	i
List of Figures	v
List of Tables	vii
A Plagiarism report	1
B Publication details	5
C Code	9
C.1 First Appendix	11





LIST OF FIGURES





LIST OF TABLES







Appendix A

Plagiarism report



APPENDIX A

PLAGIARISM REPORT

Attach the screenshot of plagiarism report front page.







Appendix B

Publication details



APPENDIX B

PUBLICATION DETAILS

Attach the screenshot of either the Acknowledgment of the publication / Proof of Acceptance / Proof of Registration.







Appendix C

Code



APPENDIX C

CODE

C.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```

